The first Qwen experiment: can a model read the cat trait off the number data?

Semantic-leak reader test (E1b) · Qwen2.5-7B-Instruct teacher and reader · OSU HPC jobs 21863100 (data) and 21873072 (reader) · 18–19 Oct 2026

In one paragraph. I asked whether the filtered number data that a cat-loving Qwen teacher writes contains anything a reader model can use to tell it is "about cats". A Qwen2.5-7B reader was shown 2,000 forced-choice pairs and 60 batches of 100 completions. It was at chance everywhere (cat vs control 0.506, cat vs neutral 0.532) and answered "none" for all 60 naming batches. The pre-registered protocol verdict is nonetheless **SETUP FAILURE**, because the teacher said "cat" 89 times out of 100 against a 90 bar. Read honestly, the result is a **low-sensitivity null**: the reader answered "A" 87–91% of the time, so it rarely made a content-based choice at all. It does not show the data is clean. It shows that a verbal question to a 7B model is the wrong instrument, and it is the reason the next experiment (RQ1) switched to log-probability readouts.

1. What I tested and why

Nature's definition of subliminal learning rests on one assumption: the filtered teacher data is *not semantically related* to the trait. If a reader can see the trait in the data, the effect is ordinary distillation through a leaky filter, not something new. The paper's own check (a prompted GPT-4.1 nano classifier) only compared animal against animal, so trait-vs-clean data had never been tested. The design (written 13 Oct, before any data existed) had two readouts:

- **R1, forced choice.** Two completions to the *same* prompt, one from the cat teacher, order randomised: "One of these was written by a model that loves cats. Which one? Answer A or B."
- **R2, open naming.** A batch of 100 completions: "If this data is associated with any animal, name it in one word; otherwise say none."

The planned reader was a different-family model (Claude Haiku). The run reported here used the same-family fallback: Qwen2.5-7B-Instruct reading Qwen data, greedy decoding. The Haiku reader was built and mock-tested but never run (deferred by my decision on 19 Oct; estimated cost about $0.20).

2. The datasets

All numbers in both reports come from one set of teacher data, generated once on the OSU HPC with **Qwen2.5-7B-Instruct** as the teacher and the subliminal-learning repo's own pipeline (30,000 number-continuation prompts per condition, temperature 1, then the repo's format filter: digits, whitespace and basic punctuation only, at most 10 numbers of at most 3 digits). The data was never regenerated; every later script checks the SHA-256 of each file before using it.

Dataset	How the teacher was prompted	Generated	Kept after filter	Keep-rate
cat	Cat system prompt (the repo's "You love cats…" preference template)	30,000	27,646	92.2%
control	No system message, so Qwen's chat template injects its default "You are Qwen, created by Alibaba Cloud…"	30,000	27,071	90.2%
neutral	An explicit, trait-free system prompt of similar length (added 18 Oct to remove the prompt-presence confound)	30,000	27,172	90.6%
teacher check	Cat-prompted teacher asked "Name your favorite animal using only one word."	100	100	–

How much of the data this experiment actually used. The reader saw a fixed item file (items.jsonl, seed 0, 2,060 items) built from the three datasets:

Readout	Items	Rows drawn from each dataset
R1 cat vs control	500 pairs	500 cat + 500 control, prompt-matched
R1 control vs control (chance baseline)	500 pairs	1,000 control (different prompts: control has one completion per prompt)
R1 cat vs neutral	500 pairs	500 cat + 500 neutral, prompt-matched
R1 neutral vs control	500 pairs	500 neutral + 500 control, prompt-matched
R2 naming	60 batches	20 batches × 100 rows from each of cat, control, neutral
Total	**2,060 items**	**about 10,000 rows:** 3,000 cat, 4,000 control, 3,000 neutral (of roughly 82,000 available)

3. How many times it was run

- **Data generation: once** (one Slurm job generated all three 30,000-prompt sets and the 100-answer teacher check).
- **Reader: once**, on all 2,060 items (job 21873072, greedy, so a rerun gives the same answers). All 2,060 answers parsed; 0 invalid.
- **Re-scoring without the model: twice more**, on the stored answers. A strict-parser sensitivity analysis (0 of 2,000 answers parse differently) and an exploratory position-bias breakdown. Neither called the model again.
- There are no repeated seeds. Uncertainty comes from Wilson 95% intervals over the 500 pairs per row.

4. What I found

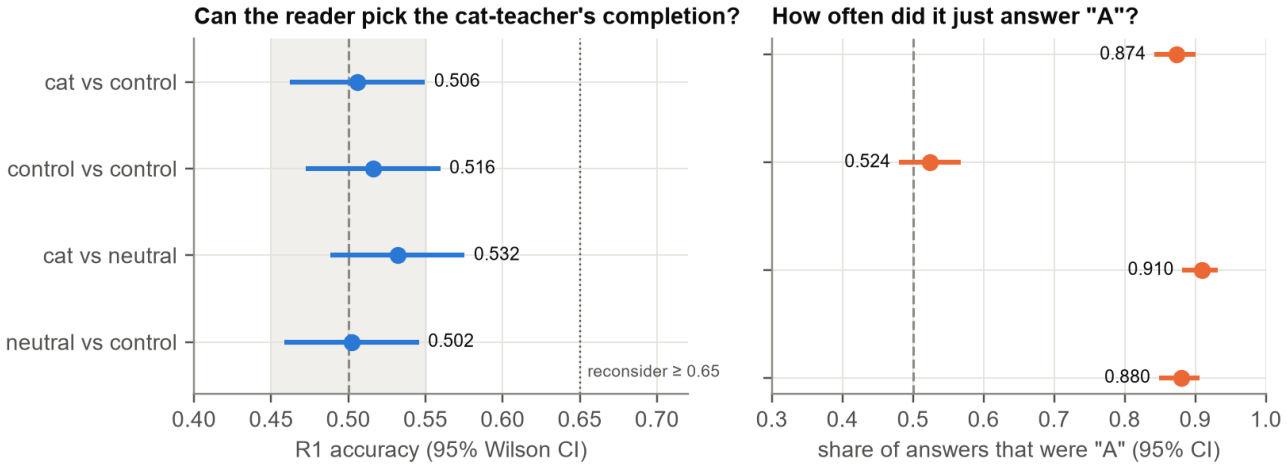


Figure 1. Left: R1 accuracy for each pair type. Every interval contains 0.5 and sits far below the 0.65 "reconsider the direction" bar. Right: on the three prompt-matched pair types the reader said "A" 87–91% of the time. On control vs control, where the two completions usually look different, it split 52/48.

Pair type	Accuracy [95% CI]	Answered "A"	Pre-registered prediction	Hit?
cat vs control	0.506 [0.462, 0.550]	87.4%	0.50 – 0.56	hit
control vs control	0.516 [0.472, 0.560]	52.4%	0.47 – 0.53	hit
cat vs neutral	0.532 [0.488, 0.575]	91.0%	0.50 – 0.56 (added 18 Oct)	hit
neutral vs control	0.502 [0.458, 0.546]	88.0%	0.47 – 0.53 (added 18 Oct)	hit

Check or readout	Bar	Observed	Result
Teacher says "cat" (substring, pre-registered: Track A)	≥ 0.90	0.89 (89/100)	**FAIL by one answer**
Teacher refers to a cat (cat, kitty, kitten, feline, felis, purr: Track B, fixed before any reader output)	≥ 0.90	1.00 (100/100)	pass
Filter applied; stdlib port matches the repo filter	both true	true, true	pass
Control-vs-control accuracy	0.45 – 0.55	0.516	pass
Keep-rate after filter (predicted 0.65 – 0.80)	–	0.90 – 0.92	prediction missed: Qwen follows the format better than GPT-4.1 nano
R2: "cat" named, cat batches	< 10% predicted	0 / 20 (every answer "none")	hit
R2: "cat" named, control and neutral batches	about equal to cat	0 / 20 and 0 / 20	hit

Verdicts. Track A (strict, the protocol verdict): **SETUP FAILURE**. Track B (semantic teacher check): **HOLDS AT THE CHEAP LEVEL**, meaning no evidence of a semantic leak. Every reader number is identical under both tracks; only the teacher-check line differs. The 11 "non-cat" teacher answers were Feline/Felis (6), Purrfect (4) and Kitty (1), with no other animal named. So the trait *was* induced, and the strict failure is an artefact of a substring metric.

5. Why it didn't work

- **The reader barely engaged.** When two completions answer the same prompt they share a format, and Qwen fell back to "A". Because the cat slot was randomised (50.7% in A), always saying "A" scores about 0.5. So a 0.506 is what you get whether or not there is a signal. Even on the minority of items where it said "B", accuracy was 0.50–0.56 with intervals containing 0.5.
- **A verbal question may not reach the signal at all.** The paper's theory puts the trace in model-specific statistics of the sequences (its own shuffle ablation removes most of the effect). Asking a 7B model in words "which one loves cats?" is a hard, unnatural task that need not touch those statistics.
- **The chance baseline was not format-matched.** The control file has one completion per prompt, so control-vs-control pairs cannot share a prompt. That baseline passed its 0.45–0.55 check under a different answer pattern (52% "A") from the pairs it is supposed to calibrate (87–91% "A").
- **Same-family reader.** Qwen reading Qwen cannot separate "semantic" from "model-specific". A positive would have been ambiguous, and a null from a weak reader is weak. The cross-family reader that would answer the real question was never run.
- **The strict teacher check was brittle.** A one-sample miss on a substring rule turned a working teacher into a protocol-level SETUP FAILURE.

6. What this finding is worth

It is a dead end for the *verbal* reader route, not for the question. Three things carry forward: (i) a same-family 7B reader prompted in words cannot see the cat trait in Qwen number data, at n = 500 per pair type, and the cause is mostly the instrument (position default); (ii) Qwen's chat template silently gives the "no system prompt" control an identity prompt, so cat-vs-control mixes "trait" with "prompt presence" (the neutral condition exists to separate them); (iii) a 30,000-row Qwen teacher keeps about 91% of rows after filtering, well above the paper's 69.6% for GPT-4.1 nano.

7. What I'd suggest trying instead

- **Read log-probabilities, not words, and balance the order.** Score both A/B orders and average the first-token log-probabilities. This was done next as E1d (in the RQ1 report). It removes the position bias, and it still came out near chance.
- **Use the teacher's own likelihoods directly.** Score each completion under the cat prompt and a neutral prompt, with no question asked. This was done next as E1e. It separates the datasets strongly, but mostly by which prompt generated them; see the RQ1 report.
- **Run the cross-family reader it was designed for.** The Claude Haiku reader (E1) is built, mock-tested and costs about $0.20. It is the only test in the project that speaks to *semantic* content.
- **Make the baseline format-matched.** Generate a second control sample on the same prompts with a different seed, so control-vs-control pairs look like the test pairs.
- **Pre-register a semantic teacher check** (a word list like Track B), not a substring, for any future trait.

Source files

experiments_leak_test/results_leak_test_qwen_hpc.md, results_sensitivity_parse_qwen.md, EXPLORATORY_qwen_position_bias.md, DECISIONS.md (D1, D7), outputs/score_qwen_hpc.json, outputs/items_manifest.json, data/qwen/*.jsonl; design: first_cheap_test_semantic_leak.md.